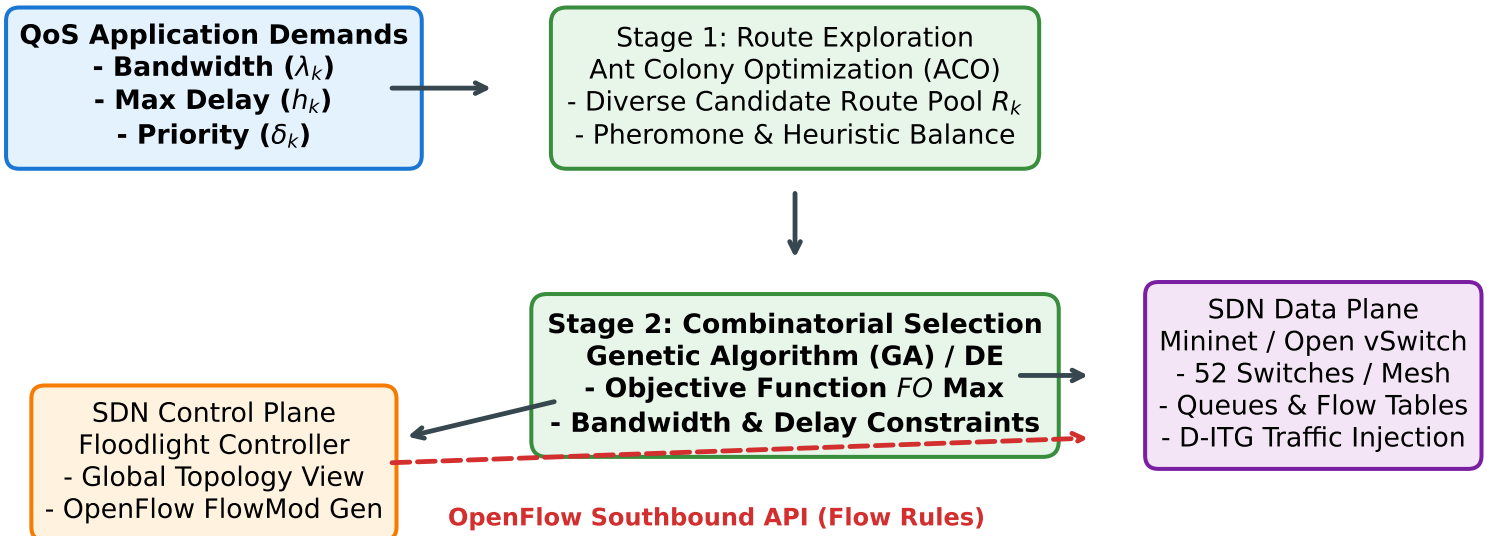
QoS Application Demands
- Bandwidth (λ_k)
- Max Delay (h_k)
- Priority (δ_k)
Stage 1: Route Exploration
Ant Colony Optimization (ACO)
- Diverse Candidate Route Pool R_k
- Pheromone & Heuristic Balance
Stage 2: Combinatorial Selection
Genetic Algorithm (GA) / DE
- Objective Function FO Max
- Bandwidth & Delay Constraints
SDN Data Plane
Mininet / Open vSwitch
- 52 Switches / Mesh
- Queues & Flow Tables
- D-ITG Traffic Injection
SDN Control Plane
Floodlight Controller
- Global Topology View
- OpenFlow FlowMod Gen
OpenFlow Southbound API (Flow Rules)